

Dushyant Baroliya

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EDUCATION

Nitte Meenakshi Institute of Technology

Bengaluru, India

Bachelor of Engineering in Mechanical Engineering, Minor in Artificial Intelligence

2023 - 2027

- Minor in Artificial Intelligence alongside a mechanical core; self-directed systems and machine-learning work carried out through the projects below

EXPERIENCE

Advanced Software Engineering (ASE) Club

Jul. 2024 - Oct. 2025

Research Analyst

Nitte Meenakshi Institute of Technology

- Organised a **national-level hackathon** hosted by the club, drawing **700+ participants** from colleges nationwide

PROJECTS

LAZR | Flutter, Dart, SQLite FTS5, Riverpod, on-device RAG

Jul. 2026 - Present

- Building a **privacy-first, on-device personal intelligence app** that indexes local files, notes, and PDFs and answers questions about them with **zero network egress** (airplane-mode guarantee, no accounts, no telemetry)
- Engineered a hybrid retrieval stack (**SQLite FTS5 keyword + vector similarity**) feeding a grounded-only answer orchestrator behind a **0.25 confidence gate**; an adversarial test suite asserts the model is never called on ungrounded queries, and every answer carries citations
- Shipped a page-anchored ingestion pipeline (txt/md/PDF extractors, chunking, embeddings) and a derived topic-clustering view; **56 tests green**, analyzer clean, Clean Architecture across capture/index/recall layers
- Verified on a **Pixel 7 device pass**: 60.6 MB release APK with **0.48–0.82 s cold start**, capture to indexed full-text recall confirmed end to end

Ledger | Live | GitHub | Rust, WebAssembly, React, TypeScript, Canvas

Jul. 2026

- Wrote a **zero-dependency price-time-priority matching engine in Rust** with integer ticks and lots, limit/market orders, **GTC/IOC/FOK** time-in-force, cancels, and a typed event stream; **19/19 tests** including a byte-identical **replay determinism** proof
- Compiled the engine to **WebAssembly** (wasm-bindgen) and streamed depth snapshots as flat Float64Array buffers to drive a **60 fps canvas order-book ladder**, print tape, and order ticket in 81 kB gzipped total
- Modeled market flow with seeded quote/aggress/cancel agents so price **emerges from the book** rather than from a price series; users trade against the simulation through the same engine path, with no backend or database by design

Signal | Live | GitHub | TypeScript, Canvas, Vite, Vitest

Jul. 2026

- Built a four-chapter **explorable explanation of options pricing**: payoff diagrams, **Black-Scholes** value vs. intrinsic, the **Greeks** as linked small multiples, and a drag-to-rotate **3D implied-volatility surface** hand-projected in canvas with no WebGL
- Drove every figure from one reactive store so a strike change in chapter 1 propagates through the Greeks in chapter 3, making the parameter coupling itself the pedagogy
- Validated the pricing math with **13 tests** against **Hull textbook values**, **put-call parity**, and finite-difference checks on each Greek; shipped at 10 kB gzipped with no UI framework

Classroom Attention Monitor | Python, FastAPI, YOLO, MediaPipe, OpenCV, React, SQLAlchemy

Jul. 2026

- Built a real-time classroom engagement pipeline: **YOLO** person/phone detection, a self-written **ByteTrack-style tracker**, **MediaPipe** FaceMesh/Pose, head pose via **solvePnP**, EAR blink detection, and hand-raise/phone association into an explainable **0–100 attention score**
- Served results over **FastAPI REST + WebSocket** into a six-route React dashboard (React Query, Recharts, TanStack Table), deriving insights client-side from existing snapshots rather than new endpoints
- Persisted sessions in SQLite via **SQLAlchemy 2 + Alembic** migrations; **106 tests** pass against a fakes-injected pipeline. Deliberately ships **no face recognition**, documented as an ethics constraint

TECHNICAL SKILLS

Languages: Rust, TypeScript, Python, Dart, SQL, JavaScript

Frameworks: React, Next.js, Flutter, FastAPI, Vite, Tailwind CSS, SQLAlchemy

ML & Vision: YOLO, MediaPipe, OpenCV, NumPy, SciPy, embeddings, vector search, RAG

Tools: Git, WebAssembly (wasm-bindgen), Docker, SQLite, Vitest, Pytest, Vercel, Android SDK/adb